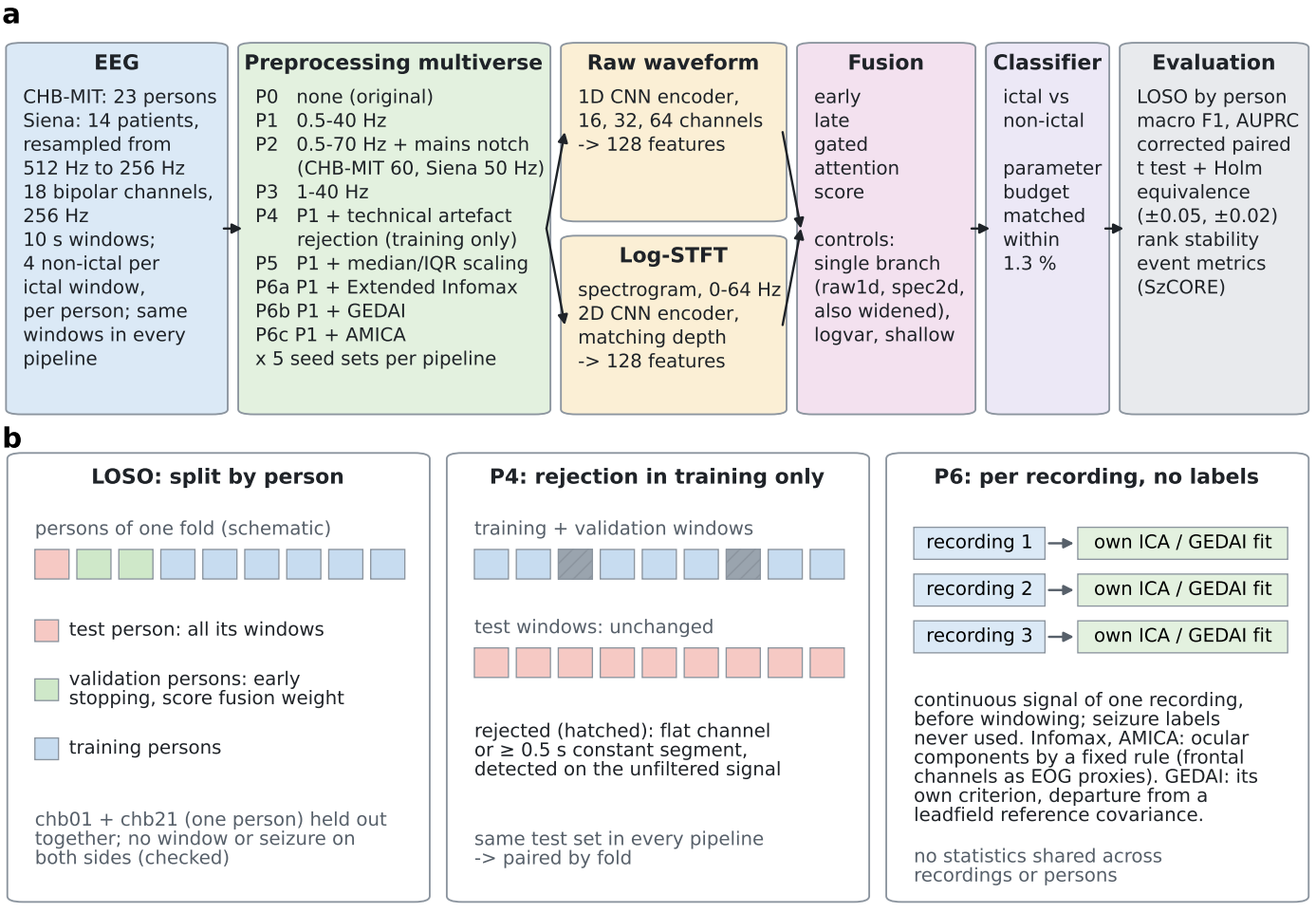
a
EEG
CHB-MIT: 23 persons
Siena: 14 patients,
resampled from
512 Hz to 256 Hz
18 bipolar channels,
256 Hz
10 s windows;
4 non-ictal per
ictal window,
per person; same
windows in every
pipeline
Preprocessing multiverse
P0 none (original)
P1 0.5-40 Hz
P2 0.5-70 Hz + mains notch
(CHB-MIT 60, Siena 50 Hz)
P3 1-40 Hz
P4 P1 + technical artefact
rejection (training only)
P5 P1 + median/IQR scaling
P6a P1 + Extended Infomax
P6b P1 + GEDAI
P6c P1 + AMICA
x 5 seed sets per pipeline
Raw waveform
1D CNN encoder,
16, 32, 64 channels
-> 128 features
Log-STFT
spectrogram, 0-64 Hz
2D CNN encoder,
matching depth
-> 128 features
Fusion
early
late
gated
attention
score
controls:
single branch
(raw1d, spec2d,
also widened),
logvar, shallow
Classifier
ictal vs
non-ictal
parameter
budget
matched
within
1.3 %
Evaluation
LOSO by person
macro F1, AUPRC
corrected paired
t test + Holm
equivalence
(±0.05, ±0.02)
rank stability
event metrics
(SzCORE)
b
LOSO: split by person
persons of one fold (schematic)
test person: all its windows
validation persons: early
stopping, score fusion weight
training persons
chb01 + chb21 (one person) held out
together; no window or seizure on
both sides (checked)
P4: rejection in training only
training + validation windows
test windows: unchanged
rejected (hatched): flat channel
or ≥ 0.5 s constant segment,
detected on the unfiltered signal
same test set in every pipeline
-> paired by fold
P6: per recording, no labels
recording 1
own ICA / GEDAI fit
recording 2
own ICA / GEDAI fit
recording 3
own ICA / GEDAI fit
continuous signal of one recording,
before windowing; seizure labels
never used. Infomax, AMICA: ocular
components by a fixed rule (frontal
channels as EOG proxies). GEDAI: its
own criterion, departure from a
leadfield reference covariance.
no statistics shared across
recordings or persons